



EXPERIENCE

Apr 2024 – Now
fulltime

Lead Software Engineer

Open Government Products

- Evolved the org's internal design system (which I built and maintained) into oui, an SSR-compatible successor with 36+ accessible components for Next.js App Router – now the default design system for newly created products in the organisation.
- Diagnosed ask.gov.sg's EOL framework (RemixJS v1) as a maintainability and open-source risk; led a full 4-month technical rewrite through influence rather than positional authority – zero regressions, with measurable improvements to performance and SEO; architecture patterns extracted into the org-wide Starter Kit, now the default starting point for newly created products in the organisation.
- Inherited maps.gov.sg as a hackathon POC with unresolved production gaps; hardened it with queuing, auth middleware, and rate limiting, sustaining 20× growth (25k→502k views) without app-level incidents; clean handoff resulted in product being promoted to a full dedicated team.
- Built an internal CLI tool to replace multi-day manual infra provisioning; generates isolated, production-ready AWS environments with CI/CD in ~10 min – adopted as the standard for new product infrastructure across the organisation.
- Embedded in the Isomer rebuild to establish engineering foundations; set up CI/CD, introduced Storybook and Chromatic for design-engineer handoff, and instituted lightweight ADRs for knowledge transfer; rebuilt components to high accessibility standards – practices fully adopted by the team and persisted after handoff.
- Drove org-wide test coverage reporting to Datadog, creating accountability for CI adoption and surfacing maintainability signals across all product teams.

Next.js / React / TypeScript / Pulumi (AWS) / Design Systems

Oct 2021 – Apr 2024
fulltime

Senior Software Engineer

Open Government Products

- Tech-led MyActiveSG+ end-to-end, advocating for unified ownership over a proposed two-team split; built a working module prototype in one day that secured internal buy-in and countered a vendor lock-in, saving ~\$12M; drove the product through two pilots to public launch – handling ~150k bookings per month at handoff.
- Led zero-downtime migration of FormSG's frontend from AngularJS to React 17, eliminating a critical maintenance liability while keeping the service live throughout.
- Authored the org's internal design system (adopted by 70% of live projects) and starter kit template (used to bootstrap 70% of all products in 2024), converting institutional best practices into durable, reusable defaults.

React / TypeScript / Node.js / Design Systems

Feb 2020 – Oct 2021
fulltime

Software Engineer

Open Government Products

- Led full TypeScript migration of FormSG's REST API, refactoring to RESTful conventions and introducing end-to-end type safety across the entire backend.
- Built the clinician-facing frontend for Singapore's national COVID-19 vaccination appointment system – the primary interface for clinic staff managing patient vaccinations at scale.

Node.js / TypeScript / React

May – Aug 2019
internship

Software Engineer Intern, Google Pay

Google Singapore

Solely scoped, designed, and shipped a new end-to-end feature for the Google Pay Android application.

Java / Android

EDUCATION

2016 – 2019

B.Comp (Hons) in Computer Science
CAP: 4.45/5.00, Honours (Distinction)

National University of Singapore